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Supplementary information

Increase in suicide following an initial decline during the COVID-19 pandemic in Japan

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Supplementary Materials for

Increase in suicide following an initial decline during the COVID-19 pandemic in Japan

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Supplementary Notes 1-5
Supplementary Figures 1-4
Supplementary Tables 1-7
Supplementary References

Supplementary Notes

Supplementary Note 1. Data sources and description

Map data of Japan (Fig. 1): The shape file of Japan is obtained from the web page of ESRI Japan inc.[1] The maps are created by authors.

COVID-19 cases in Japan (Fig. 1b): The data are obtained from the web page of the Ministry of Health, Labour and Welfare in Japan[2]. The data include confirmed cases, deaths, and recovered at the prefectural level (47 prefectures in Japan).

Suicide rate among children and adolescents, and students (Fig. 3d and Fig. 4f): We obtain the suicide rate across job status from the same sources as the main analyses[3]. We distinguish children and adolescents (individuals aged below 20), and students because some students are older than 20 years old, and some children and adolescents could be employed full-time. In 2019, the number of suicide deaths among children and adolescents was 659, while that among students was 856. These might generate some difference in the results.

Suicide rate across job status (Fig 4): We obtain the suicide rate across job status from the same sources as the main analyses[3]. The results of this analyses should be interpreted with caution because job status may not be based on the information from the subjects who committed suicide. Second, to protect privacy, the cities with one or two suicide deaths (in each month) are dropped from the observations. As a result, 5.2% of the populations are dropped. Note that, because these data are collected at more aggregated level, observations are reduced from 1,848 cities to 791 cities (some cities are counted as a greater city area as a whole).

Municipality-level socioeconomic status (Figs. 5c and Extended Data Fig. 5c): We use income and urban population to investigate whether the effects vary across different type of cities. We obtained these data from the "Statistical Observations of Municipalities", published by the Ministry of

Internal Affairs and Communications[4].

Employment and unemployment rate (Figs. 6 a and b, Supplementary Fig. 1c, and Extended Data Fig. 5b): The data are obtained from the Labour Force Survey from the Statistic Bureau of Japan, which is provided by the Ministry of Internal Affairs and Communications[5].

Domestic violence help calls (Fig. 6c): The data are obtained from the Gender Equality Bureau Cabinet Office. The data described are published by the 2nd research meetings held by the "Research Group on the Influence of COVID-19 on Women"[6].

Household income (Fig. 6d): The data are derived from the Family Income and Expenditure Survey. The survey is conducted by the Ministry of Internal Affairs and Communications (MIAC)[7], and the data include detailed information about household income, expenditure, and ownership of different facilities and appliances. We use total income after excluding bonus payment for our analyses.

Unconditional cash transfers (Fig. 6d): The data are derived from the MIAC[8]. All Japanese citizens received cash benefits at 100,000 yen (940USD) in response to the pandemic. Note that before 5 June, 2020, 30.2% of the cash benefits were distributed and the starting date is not available. We assume that it took 2 weeks to distribute the benefits in Fig. 6d.

Number of Bankruptcies (Fig. 6e): We obtain the data from Nationwide Business Failures from Tokyo Shoko Research[9].

Claims for business subsidy (Fig. 6e): We derive the data from the Ministry of Health, Labour, and Welfare[10]. The website reports the claims for a business subsidy, the number of accepted applications, and the total amount of support determined. These are reported on a weekly basis.

Working hours (Fig. 6f): The data are obtained from the Monthly Labour Survey, Ministry of Health, Labour, and Welfare[11].

Consumption Index (Supplementary Fig. 1c): We obtain the data from the Consumption Activity Index, which is provided by the Bank of Japan[12]. We use the real value of the consumption index after being adjusted for inflation.

Diffusion Index (Supplementary Fig. 1c): We obtain the data from the Indexes of Business Conditions, which is provided by the Cabinet Office[13]. We use the CI (Composite Indexes), measured as the Coincident Index.

Google Mobility Index (Supplementary Fig. 1d and Extended Data Fig. 5a): This index is published by Google by using logs of their services (e.g. Google Maps) and describes time-series trends of geographical mobility of people in a community[14].

COVID-19 cases in the world (Supplementary Fig. 3a): We obtain the data from the European Centre for Disease Prevention and Control[15], which is an agency of the European Union. The dataset includes the daily confirmed cases of COVID-19 from different countries and regions and the population. Using these two variables, we create confirmed cases per population in each country.

Stringency Index (Supplementary Fig. 3b): The data are obtained from the Coronavirus Government Response Tracker, provided by the University of Oxford[16]. We use the variable "Stringency Index" that reflects seven policy responses, including school closing, workplace closing, cancel public events, restrictions on gatherings, close public transport, stay at home requirements, and restrictions on internal movement.

Fiscal support (Supplementary Fig. 3c): We obtain the data from the Database of Fiscal Policy Responses to COVID-19 provided by the International Monetary Fund[17]. This variable reflects the critical fiscal measures that were announced or taken by each government in response to the COVID-19 outbreak as of June 12, 2020.

Weather variables: For a robustness check, we include temperature, temperature squared,

precipitation, and precipitation squared as the control variables. The data are obtained from the Meteorological Agency of Japan[18]. We use weather data from 909 stations, which do not have any missing values on temperature and precipitation during our study period. Using the six monitoring stations that are closest to the population centre in each city, and the inverse of the distance from the population centre as the weights, we aggregate the data from station-level to city-level. The weights are inversely proportional to squared distance so that closer stations are given higher weights.

Population: We use the population to construct suicide rates and to weight all the regressions. We derive the data from the Population Estimates, managed by the Ministry of Internal Affairs and Communications (MIAC)[19]. We use population measured in 2018 across gender and age cohorts (children and adolescents: age 0~14, adults: age 15~64, and elderly: age over 65). Because of data limitations, the age classification is slightly different from the suicide data.

Supplementary Note 2. Outbreaks of COVID-19 and response of the government in Japan

Until October 2020, Japan had faced two large COVID-19 outbreaks, the first mainly from February to June 2020, and the second from July 2020 and afterward. The first case of the 2019 novel coronavirus infection (COVID-19) was confirmed in January 2020 (Supplementary Fig. 1a). By February, the number of cases was increasing rapidly. Notably, the Diamond Princess cruise ship, which docked off Yokohama Bay, was hit by the virus outbreak, with about 20% of passengers and crew infected, raising concern about the disease. In response, the national government banned large-scale gatherings and requested nationwide school closure (elementary to high school) at the end of February.

In March, despite the government's measures, the virus spread did not slow down. Cases grew exponentially; for example, the total confirmed number of cases rose from 2,305 on April 1

to 10,283 on April 18. Given this, during this period, the government escalated anti-contagion policies. On April 7, it declared a state of emergency in seven cities that lasted about a month, and expanded it to all cities on April 16, in which citizens were requested to close schools (elementary to high school, and some universities delayed the start of their academic calendar), shops, and enterprises, and to stay home if possible. While there was high compliance, these actions were not mandated but requested (Fig. 1c, Supplementary Fig.1b). These nationwide efforts led to the containment of the virus. By the end of the May, in most prefectures, the daily new cases were brought down to below ten. Therefore, the state of emergency was lifted from prefectures where the infection situation was relatively mild on May 14, 2020, and nationwide on May 25, 2020.

However, another outbreak of COVID-19, which was larger than the first one, took place in the country. The number of infections began increasing again in late June 2020 and reached a peak in early August 2020 (around 1,500 confirmed cases per day). Unlike the large-scale health interventions implemented during the first outbreak, the national government was reluctant to enforce such extensive measures. Instead, the prefectural governments initiated tailored and focused health interventions to control disease hotspots. For example, the Tokyo prefectures requested that bars and restaurants shorten their business hours at the end of August, which led to a declining trend of the disease spread.

As is the case with many other countries, containment measures brought about an economic downturn: OECD estimated Japan's real GDP growth declined by 6.0% with the single-hit scenario and by 7.3% with the double-hit scenario.[20] Unemployment rates have been gradually increasing from 2.3% in January 2020 to 3.1% in October 2020.[5] The Coincident Index (one of the indexes of business conditions) has been worsening from 94.1 in January 2020 (2015=100) to 89.7 in October 2020 (Supplementary Fig. 1c). [13]

In response to the financial crisis, the government of Japan has made various subsidies and

benefits in cash and in kind available to all citizens and those who have lost their income. These include cash benefits (100,000 yen ≈ 940 USD) for all citizens; compensation for those who have lost their income for living funds and housing, and for sick leaves; increased child allowance and support for using childcare services; monetary support including the postponement of payment for social insurance; and loans without interest and security for freelance workers[21]. Among those subsidies and benefits, the Short-Time Compensation, provided for vulnerable enterprises to stabilise their employment, plays a significant role in curbing a rise in the unemployment rate. The number of claims for business subsidy drastically increased to about 1,800,000 in late October (Fig. 6).

In our analyses, we define the pandemic period as between February 2020 and October 2020, because we believe the outbreaks in Japan and the cruise ship Diamond Princess, and the subsequent government measures, could have affected the suicidal behaviours among Japanese citizens. We further classify the period between February and June as the first outbreak, and thereafter as the second outbreak. Additionally, we define the period of the state of emergency as April and May in 2020, and the school closure (elementary to high school) as March and April 2020.

Supplementary Note 3. Suicide in Japan

Overall trend: Suicide in Japan is a serious social issue, being among the top 10 causes of death for those aged between 10 and 69: notably, it is the leading cause of death among the population aged 15-39, the second for ages 40-49, the third for ages 50-54, and the fourth for ages 55-59[22]. Additionally, suicide rates in Japan have been higher than in many other high-income countries: The Japanese suicide rate is the seventh highest among high-income countries as classified by the World Bank[23] and OECD.[24]

Demography: From November 2016 to October 2020, there were 76,626 suicides with an average monthly rate of 12.8 per million population. There are large gender and age disparities in suicide rates. The suicide rate among males is about 2.3 times higher than that among females (per-month suicide rate is 18.0 per million for males and 7.8 per million for females). Particularly, males aged 20-69, who are mainly the working population, are at the highest risk of suicide (21.5 per million). A previous finding suggests that suicide deaths among males can largely be attributable to financial and economic issues. The study finds that, during the economic crisis in the 1990s, the suicide rate in Japan drastically increased, and it remained high thereafter.[25]

Although women's suicide rate is much lower than men's, it is the third-highest rate among OECD countries.[24] A study investigating 193 suicide attempts suggests that family problems or loneliness could trigger mental disorders among females, while economic issues and work-related problems are the main sources among males[26].

Suicide is one of the top causes of death among children and youth, even though those aged below 20 were at the lowest risk within the Japanese population (3.2 per million). The main causes of suicide in young generations were related to school issues. In fact, studies suggest that suicides among children and youth increase when a school session begins after a long vacation.[27] Furthermore, suicide attempts were associated with school bullying, being a sexual minority, and juvenile delinquency.[28]

Social determinants: While the reasons behind suicide are extremely complex, Japan's suicide rates have been associated with macroeconomic conditions, particularly the unemployment rate[29], and the association is stronger than in other OECD countries[30]. During the financial crisis from 1997 to 1998, major financial institutions went bankrupt, and suicide rates sharply increased. They remained high in the next decade, when the country's economy stagnated. In the 2010s, the unemployment rate started to decline (from 4.3% in 2012 to 2.4% in 2019), and the total number

of suicides also began declining. This trend continued until recently: we observe that, on average, the suicide rate declined by 6.4% during our study period (per-month suicide rate is 13.22 per million in 2017 and 12.38 in 2019).

Although economic recession could increase suicide risk, economic policies can mitigate its negative impact. A previous finding suggests that larger government expenditures were associated with a decreased suicide rate among men aged 40-64, particularly when the unemployment rate was high[31].

Other social factors, such as occupation, overwork[32], marital status, and social support[33], are reported as factors associated with suicide among the Japanese population.

Means: People took their own lives mainly at home, and the most common way of committing suicide was hanging, which comprised more than 60% of the cases and is a common method in many other countries[34]. Jumping (from a building or in front of a moving train) and charcoal-burning suicide (death by carbon monoxide poisoning) were also prevailing methods (around 10%).

Japan is remarkably different from the U.S. in its major suicide means. Thus, the concern in the U.S. that suicide deaths by firearm may increase during the COVID-19 pandemic subsequent to heightened gun sales[35,36] does not appear to be relevant to Japan.

Supplementary Note 4. Psychological response after natural disasters

We have found that suicide declined during the first COVID-19 outbreak. This may be because of positive psychological responses to the crisis, called the pulling-together effect or honeymoon phase. Existing studies often find a drop in suicide rates after national disasters, including Hurricane Katrina and the 9/11 terrorist attack. During the COVID-19 outbreak, virus containment measures required citizens to reduce their social contacts collaboratively, and this

might have generated a sense of partnership within communities. Another explanation could be that the pandemic might have altered individuals' views on life. Facing real mortal risk, people might have started thinking of life as more precious and death as more frightening, which could have prevented them from committing suicide.

The existing literature suggests that there are six phases of a disaster[37]: (1) a Pre-Disaster phase: feelings of vulnerability, worry about safety, and responsibility for subsequent negative consequences to loved ones or damage to property; (2) the Impact phase: feelings of disbelief, numbness, fear, confusion, and panic immediately after an event; (3) the Heroic phase: predominant rescue and survival behaviours for adaptation to the environment following assistance from others; (4) the Honeymoon phase: feelings of hopefulness and optimism due to community bonding arising from sharing the catastrophic experience in addition to mutual cooperation; (5) the Disillusionment phase: feelings of disappointment and resentment due to unmet emotional and physical needs because of withdrawn supports; (6) the Reconstruction phase: during the process of attempting to revive environments, some people may become resilient and strong, while heterogeneous and intertemporal responses among individuals tend to correspond to trauma level, available resources, and coping skills.

In cases of natural disasters, a systematic review[38] of the literature suggests that some studies report that suicide rates dropped in the initial post-disaster period (phase 4), while it eventually increased in the later period (phase 6). Although the COVID-19 pandemic and past natural disasters are distinctive in magnitude, our results showing a reduction and subsequent increase in the suicide rate are consistent with the prevalent patterns in previous literature (phases 4 and 5).

Supplementary Note 5. Augmented Interrupted Analysis and Different Specifications

The interrupted time-series analysis is an approach to assess the effect of large-scale intervention using longitudinal data. Because the impact of the COVID-19 pandemic is extensive, and it hit the entire country, the use of this approach seems to be reasonable. However, we think this approach could generate biased estimates for at least two reasons.

First, the pandemic's effects on the suicide rate could be confounded by seasonality. We observe that the suicide rate in February (13.0 per million) is 5.1% higher than that in January (12.4 per million) before the pandemic (2017~2019). If we adopt interrupted time-series analysis, which compares the suicide trend between 2020 January and 2020 February (before and after the first wave of the COVID-19 pandemic), the estimates might capture the seasonal variation.

Second, even if the analysis controls for the seasonality, the assumption of the interrupted time-series analysis may not be reasonable. Specifically, it assumes that 1) the suicide rate changes immediately after the initial outbreak (or any interest of events such as lockdowns), and 2) thereafter, the slope of the time-trend is linearly altered. However, the suicide effect of the pandemic seemingly appears in different ways. During the first outbreak, the number of confirmed cases gradually increased, and people may have incrementally realised the adverse impacts of the pandemic, making the first assumption unrealistic. Additionally, the extent of the adverse effects could vary across periods during the outbreak. For instance, people may feel more stress when there are more cases, the government implements strict regulations, and economic downturn is more severe, making the second assumption unsuitable.

Here, we propose an alternative strategy to estimate the effects of the pandemic, adopting a similar framework to the interrupted analysis. To do so, we first “residualise” the suicide rates by controlling for the city-specific time trend and seasonality[39]. Intuitively, we use the "left-over" that cannot be explained by the location-specific time trend and seasonality, rather than the real

suicide rate. This can be written as:

$$Y_{iym} = \exp[\mu_{iym} + \gamma_{iy}] * \varepsilon_{iym} \quad \text{(S1)}$$

where μ_{iym} denotes the city-specific time-trend and γ_{iy} represents the city-specific month fixed effects. The residual (e_{iym}) is computed as the difference between the fitted value (counterfactual) and realised value (Supplementary Fig. 4a).

Using the residual, we can try the interrupted analysis. Note that we keep the panel structure of the data instead of the longitudinal design.

$$e_{iym} = \beta * Post_{it} + \alpha_{pre.slope} * Pre_{it} + \alpha_{post,slope} * Post_{it} + \varepsilon_{iym} \quad \text{(S2)}$$

We keep the data before and after the 9 months of the COVID-19 outbreak (May 2019 to October 2020). Pre_{it} takes 1 if it is before the pandemic (May 2019 to January 2020), and $Post_{it}$ takes 1 if it is during the pandemic (February 2020 to October 2020). β measures the immediate effects and the $\alpha_{pre.slope}$ and $\alpha_{post,slope}$ represents the linear slope of the time trend during the pre-intervention and the post-intervention.

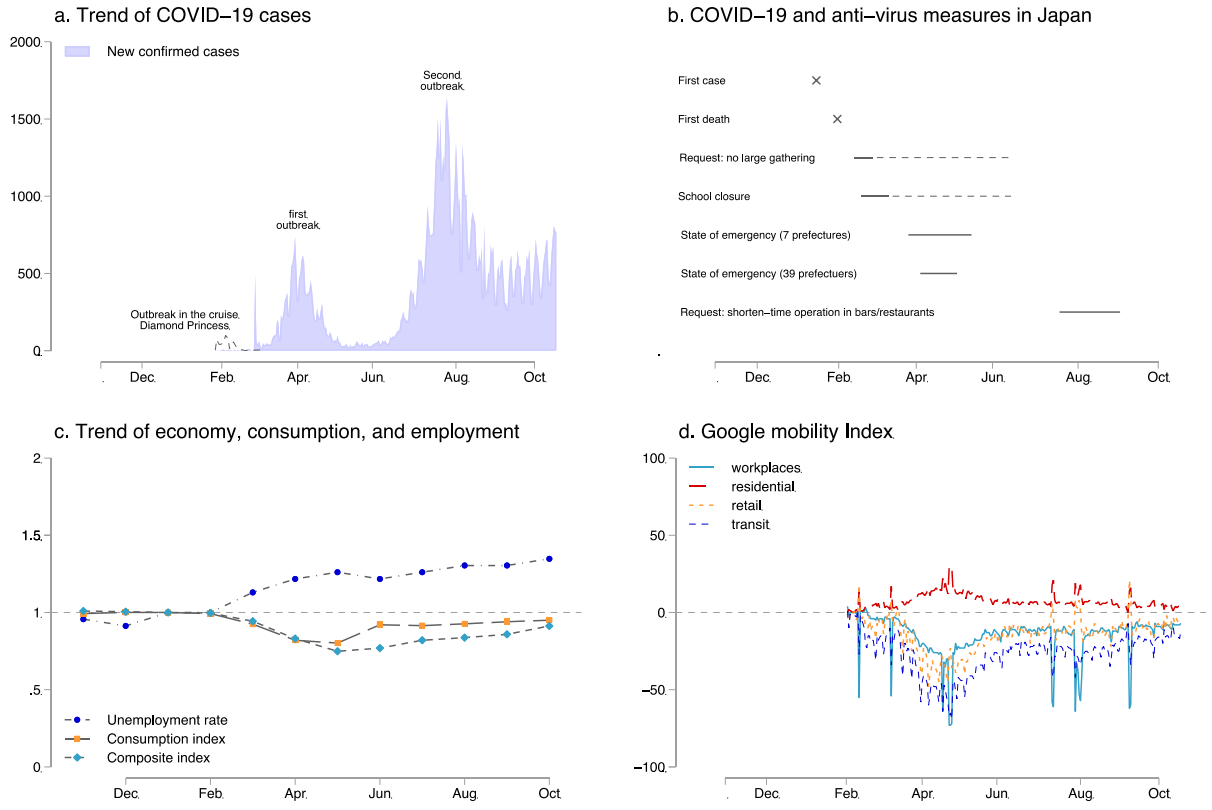
The results show that the suicide rate declined by 2.52 points (19.8%) immediately after the intervention, and the slope becomes 0.58 points (4.6%) from -0.002 points (Supplementary Fig. 4b). However, again, it may be challenging to interpret these coefficients because the effects cannot be expressed as immediate effects (β) and time trend effects ($\alpha_{post,slope}$-$\alpha_{pre.slope}$).

Alternatively, we estimate

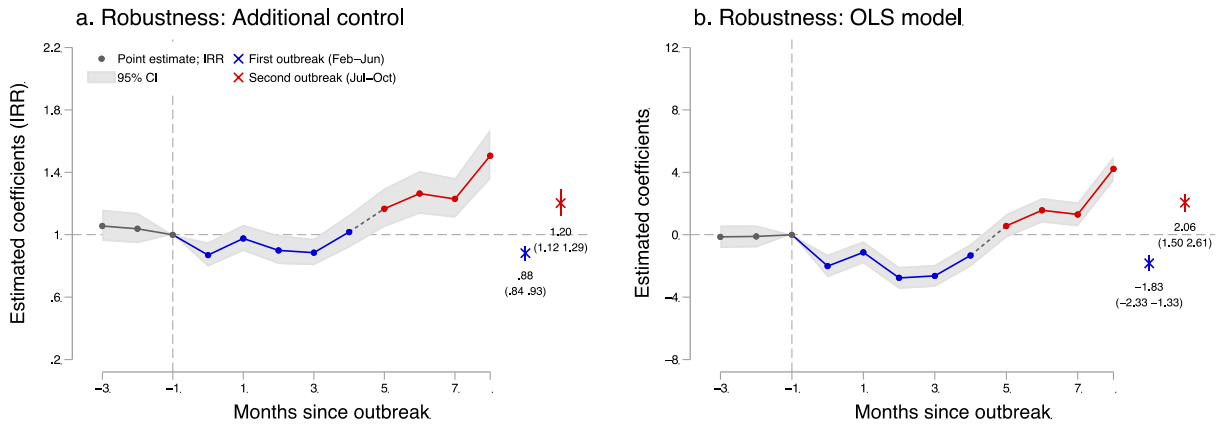
$$e_{iym} = \alpha_f * First_m + \alpha_s * Second_m + \varepsilon_{iym} \quad \text{(S3)}$$

where, $First_m$ and $Second_m$ are analogous to equation (2). In this specification, we non-linearly measure the relative level of the residual compared to the pre-pandemic periods. We find that the suicide rate declined by 0.96 points (7.56%) during the first outbreak and increased by 2.09 points (16.4%) during the second outbreak (Supplementary Fig. 4c). These results are quantitively similar to our main results.

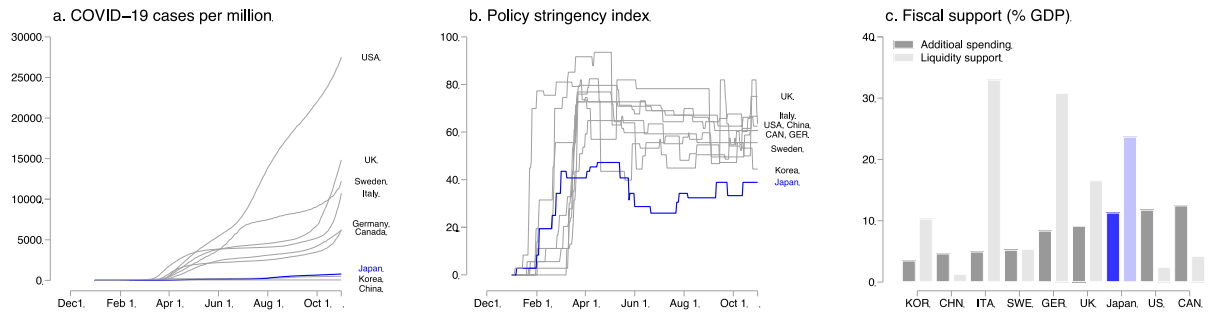
Supplementary Figures



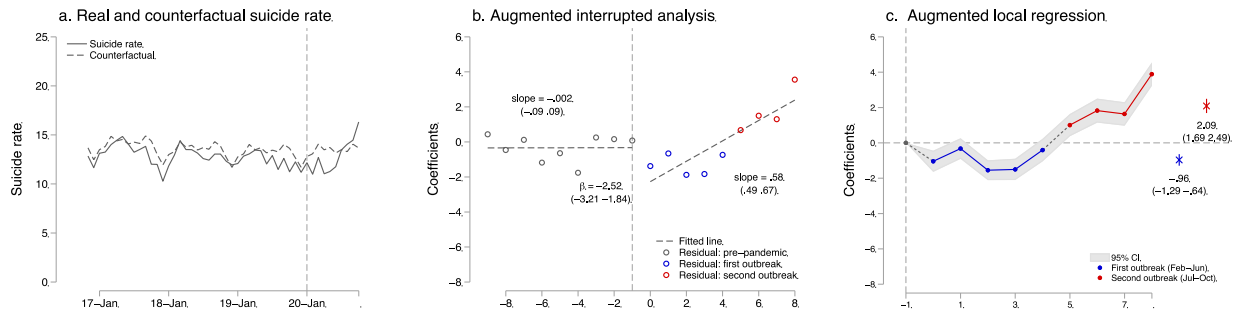
Supplementary Fig. 1. COVID-19, the virus containment measures, macroeconomic conditions, and human movement: Panel a describes daily confirmed new COVID-19 cases. We also present the outbreak on the cruise ship Diamond Princess because it docked off Yokohama Bay. **Panel b** represents the major dates of the pandemic outbreak and the government's response to the disease outbreak. **Panel c** shows the macroeconomic conditions, including unemployment rate, consumption index, and the composite index. (Jan 2020 = 1.00). **Panel d** draws the Google mobility index relative to 15 February 2020.



Supplementary Fig. 2. Robustness check: Panel a describes the results when including the time-varying controls and prefecture-specific quadratic trend. The time-varying controls include monthly average temperature and precipitation and their squares. The results are comparable. **Panel b** uses the suicide rate and the OLS instead of the Poisson model, which shows the similar trend of suicide rate. All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. N= 61,209 (a), and 88,512 (b). The separated observations are dropped for Poisson model regression (see Methods)



Supplementary Fig. 3. International comparison of COVID-19 cases, health interventions, and fiscal policies: Panels a, b, and c document the confirmed COVID-19 cases per population in different countries (panel a), the stringency index of the virus containment health intervention (b), and fiscal support as a percentage share of GDP (additional spending and liquidity support) (c). The data sources are described in Supplementary Note 1.



Supplementary Fig. 4. Robustness checks and time interrupted analysis: Panel a represents real suicide rate and predicted suicide rate estimated by the city-specific time-trend and the city-specific month fixed effects (Counterfactual). **Panel b** shows the result of the augmented interrupted analysis by using the residual, assuming that the pandemic effects are revealed as the immediate impact and slope change. **Panel c** represents relative levels of the residual compared to the pre-pandemic periods, distinguishing between the first and second outbreaks.

Supplementary Tables

Supplementary Table 1. Summary statistics

Suicide rate (per million)	All Periods (Nov 2016-Oct 2020)	Before the Pandemic (Nov 2016-Jan 2020)	First Outbreak (Feb-Jun 2020)	Second Outbreak (Jul-Oct 2020)
All	12.7	12.8	11.6	14.6
	(17.1)	(17.1)	(15.1)	(19.3)
Panel a: by gender				
Male	18.0	18.1	16.4	18.9
	(33.7)	(34.5)	(26.5)	(31.4)
Female	7.8	7.8	7.0	10.5
	(19.2)	(19.8)	(14.9)	(17.8)
Panel b: by age				
Age below 20	3.2	3.1	3.2	4.8
	(21.6)	(20.4)	(22.8)	(30.0)
Age 20-69	15.2	15.4	13.5	17.0
	(25.4)	(25.0)	(22.7)	(31.5)
Age over 70	12.1	12.1	11.5	14.0
	(27.8)	(28.1)	(25.1)	(27.5)
Panel c: by job status				
Self-employed	0.45	0.46	0.43	0.4
	(1.69)	(1.73)	(1.61)	(1.5)
Employed	2.30	2.31	2.03	3.0
	(4.24)	(4.26)	(3.92)	(4.8)
Unemployed	1.85	1.88	1.69	2.0
	(3.65)	(3.71)	(3.18)	(3.7)
Housewife	0.43	0.43	0.36	0.6
	(1.39)	(1.38)	(1.30)	(1.8)
Retired	1.91	1.96	1.65	2.1
	(3.96)	(4.02)	(3.60)	(4.0)

Notes: This table shows the mean and standard deviation (in the brackets) of each variable. The results are weighted by population. We include 1,848 cities. For gender and age groups, we use the cohort-specific population to compute the suicide rate, while we use the total population for suicide rate across different job status.

Supplementary Table 2. Full results of the effects of COVID-19 pandemic on suicide across gender and age groups using DID

	Suicide rate					
	All	Male	Female	Age Below 20	Age 20~69	Age Above 70
	(1)	(2)	(3)	(4)	(5)	(6)
Treat * First outbreak	0.86	0.86	0.82	0.98	0.85	0.87
	(0.82 - 0.90)	(0.82 - 0.91)	(0.76 - 0.89)	(0.75 - 1.27)	(0.80 - 0.89)	(0.79 - 0.95)
Treat * Second outbreak	1.16	1.07	1.37	1.49	1.17	1.11
	(1.11 - 1.21)	(1.01 - 1.13)	(1.26 - 1.49)	(1.12 - 1.98)	(1.11 - 1.24)	(1.01 - 1.22)
City-by-year Fixed Effect	YES	YES	YES	YES	YES	YES
City-by-month Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	61209	54583	35712	1896	53164	34703

Each column represents each regression, reporting incidence rate ratios and 95% confidence interval in parentheses. All regressions include city-by-year fixed effects and city-by-month fixed effects. The results correspond to Fig. 2. For age below 20, the data is aggregated at prefectural level. All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. The separated observations are dropped (see Methods).

Supplementary Table 3. Full results of the effects of COVID-19 pandemic on suicide across gender and age groups in each month using DID

	Suicide rate					
	All	Male	Female	Age Below 20	Age 20~69	Age Above 70
	(1)	(2)	(3)	(4)	(5)	(6)
Treat * Feb	0.84	0.87	0.77	0.90	0.84	0.86
	(0.78 - 0.91)	(0.80 - 0.95)	(0.67 - 0.89)	(0.59 - 1.35)	(0.77 - 0.92)	(0.72 - 1.02)
Treat * Mar	0.92	0.92	0.90	1.11	0.93	0.84
	(0.86 - 0.98)	(0.85 - 1.00)	(0.80 - 1.02)	(0.74 - 1.66)	(0.86 - 1.01)	(0.73 - 0.97)
Treat * Apr	0.80	0.82	0.74	0.77	0.79	0.85
	(0.75 - 0.86)	(0.76 - 0.90)	(0.64 - 0.84)	(0.54 - 1.08)	(0.72 - 0.86)	(0.73 - 0.98)
Treat * May	0.81	0.81	0.78	0.90	0.78	0.88
	(0.76 - 0.87)	(0.74 - 0.88)	(0.69 - 0.89)	(0.56 - 1.43)	(0.72 - 0.85)	(0.76 - 1.01)
Treat * Jun	0.91	0.89	0.93	1.24	0.89	0.91
	(0.85 - 0.98)	(0.81 - 0.97)	(0.81 - 1.05)	(0.85 - 1.82)	(0.81 - 0.97)	(0.79 - 1.05)
Treat * Jul	1.05	0.99	1.18	1.17	1.03	1.08
	(0.98 - 1.13)	(0.91 - 1.07)	(1.05 - 1.33)	(0.74 - 1.85)	(0.95 - 1.12)	(0.94 - 1.24)
Treat * Aug	1.14	1.08	1.27	1.77	1.14	1.05
	(1.06 - 1.22)	(1.00 - 1.18)	(1.12 - 1.44)	(1.21 - 2.60)	(1.05 - 1.24)	(0.91 - 1.20)
Treat * Sep	1.10	1.04	1.26	1.49	1.11	1.05
	(1.03 - 1.18)	(0.95 - 1.12)	(1.12 - 1.42)	(1.11 - 2.00)	(1.03 - 1.21)	(0.91 - 1.22)
Treat * Oct	1.36	1.18	1.82	1.50	1.42	1.28
	(1.28 - 1.45)	(1.09 - 1.28)	(1.62 - 2.04)	(1.00 - 2.25)	(1.32 - 1.53)	(1.11 - 1.47)
City-by-year Fixed Effect	YES	YES	YES	YES	YES	YES
City-by-month Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	61209	54583	35712	1896	53164	34703

Each column represents each regression, reporting incidence rate ratios and 95% confidence intervals in parentheses. For age below 20, the data is aggregated at prefectural level. All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. The separated observations are dropped (see Methods).

Supplementary Table 4. Full results of the effects of COVID-19 pandemic on suicide across gender and age groups using DID

	Suicide rate					
	All	Male	Female	Age Below 20	Age 20~69	Age Above 70
	(1)	(2)	(3)	(4)	(5)	(6)
FD2	1.02	1.01	1.05	1.01	1.05	0.94
	(0.94 - 1.11)	(0.90 - 1.12)	(0.89 - 1.24)	(0.63 - 1.63)	(0.95 - 1.17)	(0.79 - 1.12)
FD1	1.01	1.00	1.03	0.89	1.04	0.95
	(0.92 - 1.10)	(0.90 - 1.12)	(0.88 - 1.22)	(0.59 - 1.34)	(0.94 - 1.15)	(0.78 - 1.15)
D1	0.85	0.87	0.79	0.87	0.87	0.82
	(0.78 - 0.93)	(0.78 - 0.97)	(0.67 - 0.94)	(0.55 - 1.37)	(0.78 - 0.96)	(0.67 - 1.01)
D2	0.93	0.93	0.93	1.08	0.96	0.81
	(0.86 - 1.01)	(0.84 - 1.02)	(0.80 - 1.08)	(0.69 - 1.67)	(0.87 - 1.06)	(0.68 - 0.97)
D3	0.81	0.83	0.76	0.74	0.81	0.82
	(0.75 - 0.88)	(0.74 - 0.91)	(0.65 - 0.89)	(0.50 - 1.11)	(0.73 - 0.90)	(0.68 - 0.97)
D4	0.82	0.81	0.80	0.87	0.80	0.84
	(0.75 - 0.89)	(0.73 - 0.90)	(0.69 - 0.93)	(0.54 - 1.41)	(0.73 - 0.89)	(0.71 - 1.00)
D5	0.92	0.89	0.95	1.21	0.92	0.87
	(0.84 - 1.01)	(0.80 - 1.00)	(0.81 - 1.11)	(0.78 - 1.88)	(0.82 - 1.02)	(0.74 - 1.04)
D6	1.06	0.99	1.22	1.14	1.06	1.04
	(0.98 - 1.15)	(0.90 - 1.10)	(1.05 - 1.41)	(0.72 - 1.80)	(0.96 - 1.17)	(0.88 - 1.24)
D7	1.15	1.09	1.30	1.72	1.18	1.01
	(1.05 - 1.25)	(0.98 - 1.21)	(1.12 - 1.52)	(1.16 - 2.55)	(1.06 - 1.31)	(0.85 - 1.20)
D8	1.11	1.04	1.29	1.45	1.15	1.01
	(1.02 - 1.21)	(0.94 - 1.15)	(1.11 - 1.50)	(1.06 - 1.98)	(1.04 - 1.27)	(0.85 - 1.21)
D9	1.38	1.18	1.87	1.46	1.46	1.23
	(1.27 - 1.49)	(1.07 - 1.31)	(1.61 - 2.16)	(0.94 - 2.27)	(1.33 - 1.61)	(1.04 - 1.46)
City-by-year Fixed Effect	YES	YES	YES	YES	YES	YES
City-by-month Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	61209	54583	35712	1896	53164	34703

Each column represents each regression, reporting incidence rate ratios and 95% confidence intervals in parentheses. All regressions include city-by-year fixed effects and city-by-month fixed effects. The results correspond to Fig. 2. All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. The separated observations are dropped (see Methods).

Supplementary Table 5. Full results of the heterogeneous effects of the COVID-19 pandemic among age groups and gender, before and after the state of emergency and the school closure

Dependent variable:	All	Male adult	Female adult	Male elderly	Female elderly	Age Below 20
Suicide rate	(1)	(2)	(3)	(4)	(5)	(6)
Treat * COVID-19 outbreak	0.99					
(baseline in Fig 4)	(0.95 - 1.03)					
Treat * Statement of Emergency		0.79	0.73	0.85	0.80	
(Panel a in Fig 4)		(0.73 - 0.85)	(0.65 - 0.83)	(0.73 - 1.00)	(0.64 - 1.00)	
Treat * First outbreak, other period		0.9	0.85	0.87	0.82	
(Panel b in Fig 4)		(0.83 - 0.96)	(0.76 - 0.95)	(0.75 - 1.00)	(0.67 - 1.00)	
Treat * Second outbreak		1.07	1.44	1.03	1.27	
(Panel c in Fig 4)		(1.00 - 1.14)	(1.30 - 1.60)	(0.90 - 1.18)	(1.08 - 1.51)	
Treat * Before school closure						0.90
(Panel d in Fig 4)						(0.60 - 1.36)
Treat * During school closure						0.91
(Panel d in Fig 4)						(0.66 - 1.26)
Treat * After school closure						1.35
(Panel d in Fig 4)						(1.04 - 1.75)
City-by-year Fixed Effect	YES	YES	YES	YES	YES	YES
City-by-month Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	61209	47317	26319	24478	16531	1896

Each column represents each regression, reporting incidence rate ratios and 95% confidence intervals in parentheses. All regressions include city-by-year fixed effects and city-by-month fixed effects. The results correspond to Fig. 3. For age below 20, the data is aggregated at prefectural level. All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. The separated observations are dropped (see Methods).

Supplementary Table 6. Full results of heterogeneous effects of the COVID-19 pandemic across job status

Dependent variable: Suicide rate	Employed (Panel a) (1)	Retired (Panel b) (2)	Unemployed (Panel c) (3)	Self-employed (Panel d) (4)	Housewife (Panel e) (5)	Students (Panel f) (6)
Treat * First outbreak	0.80 (0.70 - 0.91)	0.80 (0.69 - 0.92)	0.83 (0.73 - 0.95)	0.81 (0.61 - 1.09)	1.17 (0.81 - 1.70)	
Treat * Second outbreak	1.24 (1.11 - 1.40)	1.08 (0.94 - 1.24)	1.11 (0.95 - 1.29)	0.95 (0.68 - 1.33)	2.32 (1.65 - 3.26)	
Treat * Before school closure						0.98 (0.61 - 1.57)
Treat * School closure						0.51 (0.34 - 0.77)
Treat * After school closure						1.05 (0.77 - 1.43)
City-by-year Fixed Effect	YES	YES	YES	YES	YES	YES
City-by-month Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	10723	9146	9246	4124	3854	3220

Each column represents each regression, reporting incidence rate ratios and 95% confidence intervals in parentheses. All regressions include city-by-year fixed effects and city-by-month fixed effects. The results correspond to Fig. 4. For age below 20, the data is aggregated at prefectural level. All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. The separated observations are dropped (see Methods).

Supplementary Table 7. Full results of heterogeneous effects of the COVID-19 pandemic across geography

Dependent variable: Suicide rate	Base suicide rate (Fig 5a)		COVID-19 cases per capita (Fig 5b)		Base income per capita (Fig 5c)	
	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)
Treat * First outbreak	0.98	0.74	0.84	0.86	0.86	0.86
	(0.92 - 1.04)	(0.70 - 0.79)	(0.78 - 0.91)	(0.81 - 0.91)	(0.79 - 0.95)	(0.81 - 0.90)
Treat * Second outbreak	1.37	0.96	1.13	1.17	1.11	1.18
	(1.29 - 1.45)	(0.90 - 1.03)	(1.04 - 1.23)	(1.11 - 1.24)	(1.00 - 1.22)	(1.12 - 1.24)
City-by-year Fixed Effect	YES	YES	YES	YES	YES	YES
City-by-month Fixed Effect	YES	YES	YES	YES	YES	YES
Obs.	30134	31075	24246	36963	26166	35043

Each column represents each regression, reporting incidence rate ratios and 95% confidence intervals in parentheses. All regressions include city-by-year fixed effects and city-by-month fixed effects. The results correspond to Fig. 5. All regressions are weighted by population, and standard errors are clustered at the city level. If the variable in a city or a prefecture is lower than the median, the city or the prefecture is defined as "low". All regressions include city-by-year fixed effects and city-by-month fixed effects and are weighted by population, and standard errors are clustered at the city level. The separated observations are dropped (see Methods).

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